UNIVERSITY OF MORATUWA

IS4990 Comprehensive Group Project

PROJECT PROPOSAL

SWOT Tank Water Tracker for Sri Lanka

An AI system that estimates how much water each tank holds, using free satellite data

Group 6 · JARVIS

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1 Introduction

Sri Lankan farming runs on water stored in tanks and reservoirs, but storage figures are reported mainly for major reservoirs, not small village tanks. In 2018 FAO named the dry zone's cascaded tank-village system (ellanga gammana) a Globally Important Agricultural Heritage System (Daily FT). The Irrigation Department reports storage for its major reservoirs, such as 75% total capacity in December 2023 (Daily FT).

This project builds an AI system that estimates how much water each tank holds, using free satellite data. It combines water area, mapped by deep learning from Sentinel-1 and Sentinel-2, with water height from SWOT, a NASA and French space agency (CNES) satellite launched in December 2022.

Farmers and irrigation officials can then see storage for every tank in a cascade and plan the next Yala or Maha crop.

2 Problem Statement

Satellites already show where tank water is, but not how much water each tank holds.

- **Area is not volume.** IFAD's 2026 maps track the water surface of more than 10,650 water bodies in six districts at 10 m resolution, but not their volume (IFAD). A shallow tank and a deep tank of the same area hold very different amounts of water.
- **Water height used to be out of reach.** Earlier altimeter satellites measured only along the line under their path; SWOT's wide swath covers far more ground (AVISO) and is specified for lakes down to 250 m × 250 m (NASA ARSET).
- **Clouds and water plants hide the water.** Optical images fail under monsoon clouds, and floating plants cover water surfaces; in North Bolgoda Lake water hyacinth spread most from May to November (Dawalagala et al., 2023).
- **No tool joins these for Sri Lanka.** No study found combines SWOT water heights with AI water maps to give storage for Sri Lankan tanks.

3 Research Questions and Objectives

The project tests whether free satellite data and AI can report water storage for small village tanks, not just the major reservoirs.

Research questions

1. How accurately can deep learning map tank water from Sentinel-1 and Sentinel-2, including tanks under cloud or partly covered by water plants?
2. How accurately does SWOT measure water height on major reservoirs and on village tanks close to its 6.25 ha size limit, compared with gauge readings, and how many "suspect" readings can a learned screen keep, and at what error?
3. Can a model trained on about 100 gauged reservoirs turn a tank's SWOT height and AI water area into % full, for tanks it has never seen, more accurately than one area–storage formula?

4. Can a deep-learning model forecast each tank's storage 2–4 weeks ahead, from recent storage, rainfall and evaporation, better than a "no change" forecast?

Main objective: build and validate a system that reports storage for five major reservoirs and eight village tanks near Anuradhapura, using only free satellite data.

Specific objectives

- Build a labelled dataset of Sri Lankan tank water masks, including cloudy and plant-covered cases.
- Train a deep-learning model that maps water through clouds and under floating plants.
- Process SWOT water heights for the study tanks, check them against gauge readings, and train a screen that decides which readings to keep.
- Train a model on about 100 gauged reservoirs that turns SWOT height and water area into % full for any tank; then estimate each study tank's storage over time and train a model that forecasts it 2–4 weeks ahead.
- Deliver a dashboard showing storage for each study tank.

4 Literature Review and Research Gap

SWOT storage has been shown on large US reservoirs and, combined with other satellites' water area, on 245 reservoirs in five countries; SWOT heights have been checked on Taiwanese ponds, while Sri Lankan work maps water area, or water level from ICESat-2 only every 91 days. We found no study combining them for Sri Lankan tanks; the closest, a 2026 Thai abstract, maps water and floating plants with Sentinel-1 and Sentinel-2 but uses no SWOT.

Table 1. Related studies and the gap each leaves for this project

Study	Where	What it does	Gap for this project
Ma et al., 2026	12 reservoirs, western USA	Uses SWOT height and area together to estimate storage; height error under 20 cm, storage error under 10%	Large US reservoirs; not small tanks, not Sri Lanka
Das and Hossain, 2025	245 reservoirs in the USA, Spain, India, South Africa and Thailand	Joins SWOT storage with Sentinel-1, Sentinel-2 and Landsat water area, using a Random Forest to calibrate the non-SWOT data, to track storage change	Not deep learning; not Sri Lanka
Wang and Tseng, 2025	14 reservoirs and 15 ponds, Taiwan	Checks SWOT heights against gauges; a refined workflow cut errors to 0.15–0.53 m and detected ponds of about 110 × 110 m	Heights only; no volume, no AI
IFAD and ESA, 2026	Six districts, Sri Lanka	Monthly 10 m water-surface maps of more than 10,650 tanks, 2017 to August 2025	Area only; no volume

Study	Where	What it does	Gap for this project
Wu et al., 2023	Four small reservoirs, California	A ResNet34 model reads water level from Sentinel-1 and Sentinel-2 images; error 0.19–0.77 m	Needs gauge records to train for every reservoir; no SWOT
Madhurshan et al., 2024 (UoM)	Victoria reservoir	Compares water area from Landsat 8, Sentinel-1 and Sentinel-2	One reservoir; area only
Meyen, 2018 (UoM MSc)	Kotmale reservoir	Forecasts water level with LSTM and GRU from gauge history and weather data	Needs gauges; no satellite
Jayawardena et al., 2021 (UoM)	Sri Lanka	Extracts water from Landsat 8 with classic classifiers; kappa above 0.95	Classic methods on Landsat; area only
Pruekthikanee, 2026	Chi River basin, Thailand	A U-Net on Sentinel-1, Sentinel-2, NDWI and NDVI separates open water from floating vegetation and flooded farmland, then builds a stage–area relationship checked against a gauge	Conference abstract with no results yet; no SWOT, no tanks
Perera and Jayaweera, 2020	Kaudulla, Senanayake Samudraya and Lunugamwehera, Sri Lanka	Compares Sentinel-1 water area from Google Earth Engine with reported volumes, 2017–2019; R^2 0.73, 0.94 and 0.67. Sentinel-2 water index failed under cloud	Area only; no water height, no curve
Madumalsha et al., 2025	Nine major reservoirs in Sri Lanka, including Nachchaduwa and Nuwara Wewa	Checks ICESat-2 laser water levels against gauges; correlation above 0.99, with an average offset of 0.37 m (satellite above gauge)	Levels only, every 91 days; no area or volume

This project's contribution: the first published SWOT storage study for Sri Lankan tanks, checked against local gauge readings, plus an AI water map that handles clouds and water plants and updates storage every 5–12 days. SWOT's own lake product already reports storage change for listed lakes; this project adds the AI maps, small tanks and ground checks.

5 Study Area and Data Sources

The study covers 13 tanks east of Anuradhapura: five major reservoirs with daily gauges and eight village tanks within about 5 km of Mahakanadarawa. They were chosen after the October 2026 data check described in the Feasibility Check section below.

Why these tanks

- **Seen by SWOT:** all 13 lie east of the blind strip of SWOT pass 273, and passes 273 and 092 both cover them, so each is seen twice every 21 days.
- **Listed by SWOT:** all 13 have a lake ID in SWOT's Prior Lake Database, so their heights come straight from the Hydrocron API, with 47–90 usable readings each since July 2023.
- **Ground truth:** the five major reservoirs have daily Irrigation Department gauge readings. They give firm results and show how far SWOT's "suspect" readings can be trusted.

- **A range of village tanks:** 49 village tanks of 6.25–25 ha near the five reservoirs pass the SWOT checks, and Mahakanadarawa has the most nearby (19). The eight chosen lie 0.4–5.1 km from it: two with open water, three mixed and three mostly covered by plants. Tanks 1 and 2 are close to SWOT's 6.25 ha (250 m × 250 m) size limit, so they test it.
- **Cascade check (29 October 2026):** the eight village tanks do not form one cascade. The Irrigation Department's catchment map and a downhill flow trace on the Copernicus elevation model agree: tanks 1–5 each drain straight into Mahakanadarawa, tank 6 drains northward, tank 7 into Nuwara Wewa and tank 8 into Kallanchiya. RQ1–RQ4 are answered tank by tank, so this does not change them. The tanks' local names are still to be confirmed with the Agrarian Services Centre; most have no name in OpenStreetMap.



Figure 1. Study area east of Anuradhapura (80.40–80.80°E, 8.10–8.60°N): the five major reservoirs and the eight village tanks.

Table 2. The 13 study tanks

Tank	SWOT lake ID	SWOT area (ha)	Usable SWOT readings	Notes
Nachchaduwa	4580051323	1,311	76 (55 best)	Major reservoir, daily gauge
Hurulu Wewa	4580013372	913	77 (71 best)	Major reservoir, daily gauge
Nuwara Wewa	4580009523	746	62 (53 best)	Major reservoir, daily gauge
Mahakanadarawa	4580006192	550	85 (63 best)	Major reservoir, daily gauge
Manankattiya	4580013332	175	71 (57 best)	Major reservoir, daily gauge
Village tank 1	4580005172	6.5	73	0.4 km from Mahakanadarawa; open water
Village tank 2	4580010522	7.4	73	1.1 km away; open water
Village tank 3	4580006282	11.1	73	1.6 km away; mixed
Village tank 4	4580010542	11.5	86	2.6 km away; mixed
Village tank 5	4580011762	8.9	77	2.7 km away; mixed
Village tank 6	4580011422	16.0	90	4.3 km away; plant-covered
Village tank 7 (Bulankulama)	4580007392	7.1	47	4.3 km away; plant-covered
Village tank 8	4580043692	12.6	89	5.1 km away; plant-covered

SWOT area is the tank's reference area in SWOT's lake database. Readings run from July 2023 to 11 October 2026; "usable" includes readings SWOT flags as "suspect", and the village tanks have no best-quality readings. Water type comes from a clear February or March 2026 Sentinel-2 image.

Table 3. Data sources

Data	Source	What it gives	Use in the project
SWOT lake product (LakeSP)	NASA PO.DAAC; the Hydrocron API returns time series by lake ID (PO.DAAC webinar)	Water height and area per pass, 21-day repeat	Water height for volume
SWOT pixel cloud (PIXC)	NASA PO.DAAC	Raw water pixels with heights	Smaller tanks missing from the lake product (stretch)
Sentinel-1 radar	Copernicus, via Google Earth Engine	Images that see through cloud	Water mapping in the monsoon
Sentinel-2 optical	Copernicus, via Google Earth Engine	10 m colour and infrared images	Water and water-plant mapping
Training data	Sen1Floods11 (Bonafilia et al., 2020) and one part of S1S2-Water (Wieland et al., 2024)	Labelled water masks for Sentinel-1 and Sentinel-2 images	Pre-training the water model before fine-tuning on Sri Lankan tanks
Rainfall	GPM IMERG (daily) and CHIRPS (monthly checks)	Daily rainfall; in Sri Lanka IMERG is better day by day and CHIRPS month by month (Perera et al., 2026)	Storage forecasting

Data	Source	What it gives	Use in the project
Official storage	Irrigation Department Reservoir Status (its public dashboards and their data sheet), saved daily from October 2026	Daily storage of 117 reservoirs (74 major since 2020, 43 medium since 2023), and daily depth and storage of 14 Anuradhapura reservoirs, including the five study reservoirs	Validation, and training data for the trust screen, the storage model and the forecast
Water-area history	Global Water Watch archive, September 2026 (Zenodo, CC BY 4.0)	Water area of about 70 of the department's reservoirs, from Landsat and Sentinel-2	Training pairs for the storage model before the AI water maps are ready
Reservoir outlines and capacity	Irrigation Department reservoir layer (ArcGIS)	Full-supply outline, area, capacity and catchment of 258 tanks	Inputs and checks for the storage model
Elevation	Copernicus DEM (30 m), via Google Earth Engine	Land height and slope around each tank	Inputs for the storage model
Gauge readings	The team, at the eight village tanks on SWOT pass dates, where a gauge post exists	Water level in the field	Validation
Historical gauge records	Irrigation Department, July 2023 onwards (to be requested)	Daily water levels of major and medium reservoirs	Checking SWOT across its whole record

6 Feasibility Check

Before fixing the study area, we tested whether the project's data can really be built. In October 2026 we pulled every SWOT lake reading from July 2023 to 11 October 2026 for a 44 × 55 km box east of Anuradhapura, and checked it against Irrigation Department gauges, Sentinel-2 images and open training data. **Verdict: feasible, with the five changes listed below the table.**

Table 4. Results of the October 2026 feasibility check

Check	What we found	Result
1. Does SWOT list the tanks?	402 SWOT lakes in the box. 58% of mapped tanks of 6.25 ha or more have a SWOT lake ID, from 34% at 6.25–10 ha to 100% above 100 ha; 86% of Irrigation Department tanks have one. All 13 study tanks have one.	Risk, handled by the choice of tanks
2. Are the heights usable?	26,641 readings on 400 lakes. The five major reservoirs have 53–71 best-quality readings each. Village tanks get almost only "suspect" readings, but 103 of 104 tanks of 6.25–25 ha have 10 or more usable readings, with a clear yearly rise and fall (median 2.3 m).	Risk, tested in RQ2
3. Do the heights match the gauges?	SWOT reads 0.40–0.64 m above the gauges at all five reservoirs (0.47 m at three of them), with only 0.07–0.13 m scatter over 42–65 pass days each. Gauge levels come from each gauge's zero (full supply level minus full supply depth) and the department's daily storage. The gap is a fixed datum offset, removed once per reservoir; same-day readings will confirm it.	Pass, after one fixed correction per reservoir
4. Can Sentinel-2 map the water?	Only 17–22% of Sentinel-2 dates were clear over the three test tanks (0–3 a month, fewest from November to February). Water area followed SWOT height at Nuwara Wewa (Spearman 0.70, only 5 pairs) but less well at two village tanks (0.68 after dropping one bad SWOT reading, and 0.22), where plants covered 49–81% of the surface.	Risk, so Sentinel-1 is now core

Check	What we found	Result
5. Is there training data?	Sen1Floods11 (Bonafilia et al., 2020): 446 hand-labelled image tiles, 1.75 GB, free. S1S2-Water (Wieland et al., 2024): free, but 165 GB in seven parts.	Pass

A best-quality reading has quality flag 0, crossover flag 1 or less and dark water 10% or less (Köhn and Nielsen, 2026); a "suspect" reading has quality flag 1.

What changed in the plan

- **Study area:** 13 tanks that SWOT lists and sees twice every 21 days, east of the blind strip of SWOT pass 273.
- **SWOT heights:** village tanks keep their "suspect" readings after outlier screening. RQ2 tests how far such readings can be trusted, using the five gauged reservoirs, which have both kinds. SWOT's own early validation found that suspect readings raise the error only from 12 to 14 cm on lakes of 250 m × 250 m to 1 km² (Fjørtoft and Pottier, 2025).
- **Water area:** Sentinel-1 radar and the "water under plants" class are core, not extras. If area still fails on plant-covered tanks, their curve comes from Imbulana's (2022) relative depth–area–capacity equation, which needs only four or five area readings.
- **Ground truth:** the Irrigation Department's Reservoir Status page shows only the current day, so we save its data automatically every day (since October 2026) and will request daily levels since July 2023.
- **Training data:** Sen1Floods11 plus one part of S1S2-Water (33 GB at most), not all 165 GB.

Cost and tools. All data and software are free, and the scripts used for this check already download and filter SWOT, Sentinel-2 and gauge data, so the pipeline starts from working code. The only cost is travel: every study tank is within about 40 km of Anuradhapura.

7 Methodology

SWOT passes build each tank's height–area curve; after that, every new Sentinel water map converts straight into storage, so updates come every 5–12 days instead of every 21.

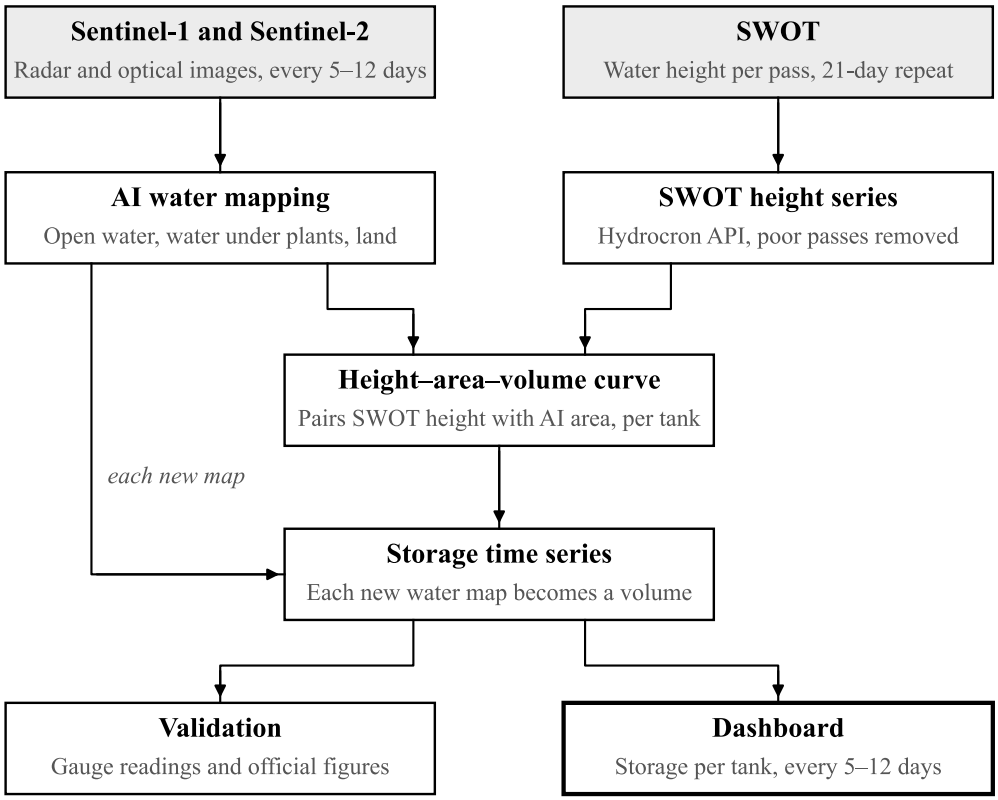


Figure 2. Method pipeline: SWOT builds each tank's height–area–volume curve; Sentinel-1 and Sentinel-2 water maps then update storage every 5–12 days.

The left path runs every 5–12 days; the SWOT path only needs enough passes, from its record since July 2023, to fit each tank's curve.

1. **Tank inventory:** record each study tank's outline, local name, SWOT lake ID and the SWOT passes that cover it; the feasibility check has already found the IDs and passes.
2. **Labelled dataset:** create water masks for many dates per tank. Start from water indices on clear Sentinel-2 images, correct them by hand, and mark three classes: open water, water under plants, and land.
3. **AI water mapping:** train a segmentation network (for example U-Net) on Sentinel-1 and Sentinel-2 together, so it still works when clouds block Sentinel-2. The tank's recent history helps it spot water hidden under plants.
4. **SWOT heights:** pull each tank's full height record since July 2023 through the Hydrocron API. Keep best-quality readings; for village tanks, which SWOT flags as "suspect", keep those too after removing outliers. Tanks without a Prior Lake Database ID need the SWOT pixel cloud instead. A trust screen (gradient boosting) then scores how wrong each reading is from SWOT's own quality fields, distance from the track, lake size and season, and decides which readings to keep. It learns from the department reservoirs with daily storage, where a reading's error is its distance from that reservoir's own height–storage curve.
5. **Storage model:** pair each gauged reservoir's daily storage, as % of capacity, with its screened SWOT height and its water area on the same day (the Global Water Watch archive first, then the AI water maps). Train gradient boosting (LightGBM) in which storage can only rise with height

and area, using only inputs every tank has: height above the tank's lowest reading, area as a share of full-supply area, full-supply area, outline shape, and catchment and slope from the Copernicus DEM. An area-only version covers tanks without SWOT. Test both on reservoirs they never saw (5 folds by reservoir), then apply them to the 13 study tanks. Each study tank also gets its own height–area–volume curve, fitted from its SWOT heights and AI water areas. Ma et al., 2026 used SWOT's simultaneous height and area the same way. This curve gives storage change above the lowest level SWOT has seen, and checks the model's % full at the village tanks.

6. **Storage time series:** after every new water map or SWOT pass, run the screen and the storage model for each study tank, and report its % full and storage change.
7. **Forecasting:** train a deep-learning model (an LSTM, with a gradient-boosting baseline) on the department's daily storage for 117 reservoirs (2020–2026), IMERG rainfall (CHIRPS for monthly checks) and ERA5 evaporation, then predict each study tank's storage 2–4 weeks ahead.
8. **Dashboard:** a map of the study tanks with each tank's current storage, trend and last update.

8 Evaluation Plan

Every result is checked against field or official data, and every AI model is compared with a simpler baseline.

Table 5. Evaluation plan

What is tested	Metric	Compared against
Water mapping	IoU and F1 for water; reported separately for cloudy and plant-covered tanks	Water-index thresholds on Sentinel-2, a Sentinel-1 threshold method, NASA's OPERA DSWx-S1 maps (which include an inundated-vegetation class, for water bodies over 3 ha and 200 m wide; NASA JPL OPERA, 2024), and IFAD/ESA maps where shared
SWOT water height	Error in cm against gauge readings on pass dates, by tank size and for best-quality versus "suspect" readings, after matching each gauge's zero to SWOT (Sri Lankan gauges read about 0.4 m below ICESat-2; Madumalsha et al., 2025). The trust screen is tested on reservoirs it never saw: the share of readings kept and their error, against SWOT's own quality flags, the hand-made filter of Levenson et al. (2025), and the cross-track filter of Lappicy et al. (2026), retuned for SWOT version D	SWOT requirement (Fjørtoft and Pottier, 2025): 10 cm (1σ) for lakes over 1 km²; goal: 25 cm (1σ) for lakes from 250 m × 250 m to 1 km² (NASA ARSET)
Storage model (RQ3)	Error in % of capacity on reservoirs it never saw (5 folds by reservoir), for all tanks and for those under 2 km², with and without the trust screen; storage change against Kanhindawewa's drone survey and the five gauged reservoirs	One area–storage formula for all tanks; area × fixed depth; Imbulana's (2022) equation where depth is known; each tank's own curve (upper limit)
Update frequency	Days between storage updates	SWOT alone (21-day repeat)
Forecast	Error at 1, 2 and 4 weeks ahead, as a % of capacity, on reservoirs and years the model never saw	"No change" and the seasonal average

What is tested	Metric	Compared against
Usefulness	Feedback from irrigation officers and farmer organisations	—

9 Expected Outcomes and Contributions

The project delivers the first published, gauge-checked SWOT storage figures for Sri Lankan tanks, plus the data and tools to repeat them.

1. **Evidence:** SWOT's accuracy on Sri Lankan tanks, measured against gauges and broken down by tank size, and a trust screen that decides which SWOT readings to keep.
2. **Model and dataset:** an AI water map that works under clouds and floating plants, with the labelled Sri Lankan tank dataset behind it.
3. **Storage model:** a model that turns SWOT height and water area into % full for tanks with no gauge or capacity table, plus a height–area–volume curve for every studied tank, reusable by irrigation engineers.
4. **Tool:** a dashboard showing storage for the 13 study tanks, updated every 5–12 days, with a 2–4 week forecast.
5. **Publication:** a paper for a conference such as UoM's ICITR.

10 Scope

The core is deliberately limited to 13 tanks near Anuradhapura, because the full idea is larger than a typical final-year project. UoM IT final-year computer vision papers usually fine-tune a pretrained model on about 600–1,250 images and add a web app (ICITR 2024), while most UoM satellite studies use one satellite and classic classifiers. This project sits above the first and close to Master's-level satellite work.

Table 6. Project scope

Level	Included	Why
Core (must deliver)	Five major reservoirs and eight village tanks; AI water mapping from Sentinel-1 and Sentinel-2 with a water-plant class; SWOT heights, including screened "suspect" readings for village tanks; the SWOT trust screen; the storage model; height–area–volume curves; storage time series; the 2–4 week storage forecast; validation; simple dashboard	Answers research questions 1–4 end to end; the feasibility check showed that Sentinel-1 and the plant class are needed, not extras
Stretch (if time allows)	SWOT pixel cloud for tanks missing from the lake product; cloud-removal module; more of the 49 village tanks that passed the SWOT checks	Wider coverage and cleaner images
Out of scope	Whole-island coverage; mobile app; real-time alerts; surveys of tank beds	Too large for 3 people in 8 months

11 Tools, Equipment and Data Access

All data and software are free; the only real cost is travel to the study area for gauge readings.

Table 7. Tools, equipment and data access

Type	Item	Purpose
Data access	NASA Earthdata account	SWOT lake data through PO.DAAC and the Hydrocron API
Data access	Google Earth Engine	Sentinel-1, Sentinel-2 and rainfall data
Software	Python, PyTorch, xarray, rasterio, geopandas	Models and satellite data processing
Software	QGIS	Checking maps and drawing tank outlines
Software	FastAPI, Next.js, Leaflet	Dashboard
Compute	Google Colab GPU and team laptops	Training and testing
Field	Team smartphones (camera and GPS)	Photographing gauge posts on SWOT pass dates

12 Team Roles

Each member owns their research questions end to end and trains one model: A the water map (RQ1), B the storage model, with the SWOT trust screen (RQ2 and RQ3), and C the storage forecast (RQ4). All three share field trips, labelling and writing.

Table 8. Team roles

Member	Owns	Key outputs	Research question
Member A: water mapping	Labelled dataset, Sentinel-1 and Sentinel-2 segmentation model, water-plant class	Water maps for every tank and date	1
Member B: SWOT and storage	SWOT processing, gauge data collection, the trust screen that cleans every reading, and the storage model (gradient boosting) that turns SWOT height and water area into % full	SWOT accuracy report by tank size and reading quality, the trained screen and storage model, and % full for every tank	2 and 3
Member C: forecast and system	Storage time series (B's model run on every new map and pass) and its validation, the storage forecast model (LSTM, with a gradient-boosting baseline), data pipeline, dashboard, user feedback	Storage time series, working dashboard, and the trained forecast model with its results	4, plus whole-system testing

13 Timeline

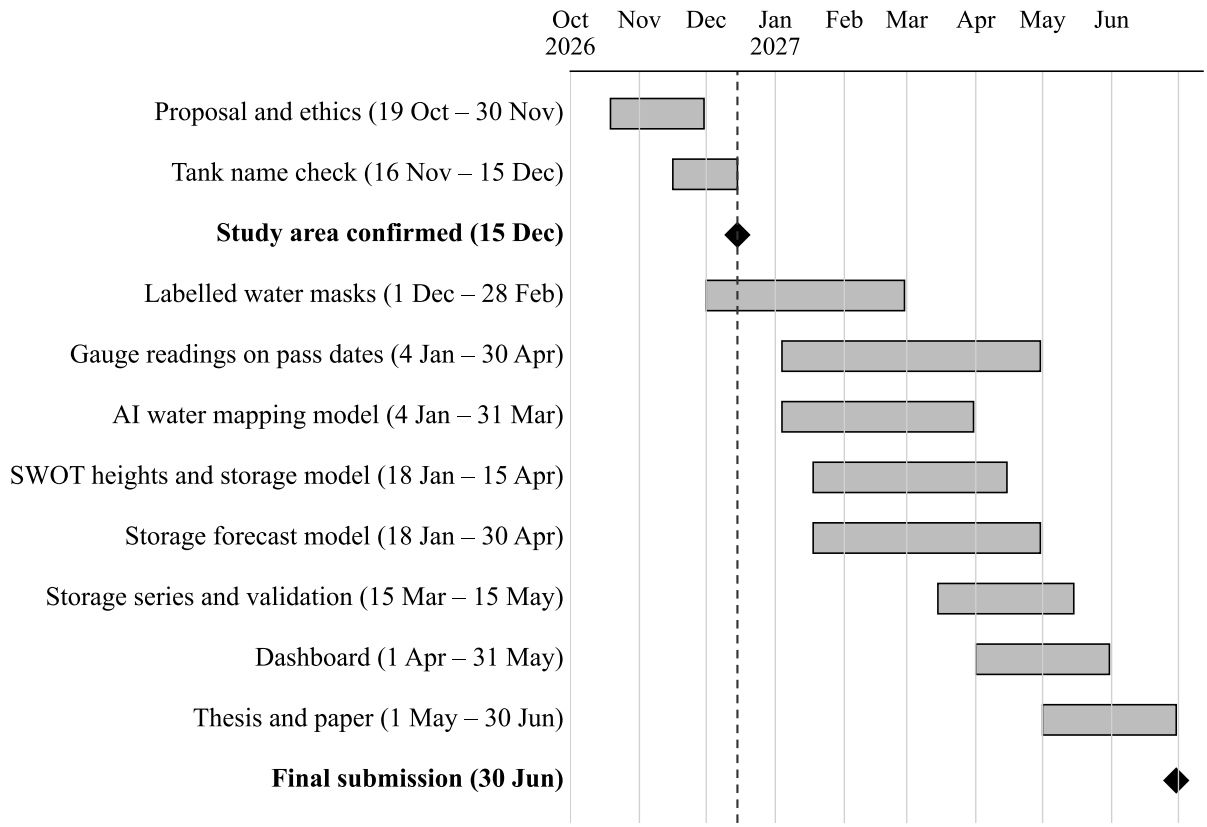


Figure 3. Project timeline, October 2026 to June 2027. Dates are assumed and will be adjusted to faculty deadlines; the study area is confirmed by 15 December, after the cascade and name checks.

Gauge readings run from January to April while the models are built, so validation data is ready when storage estimates arrive in March. Historical gauge records since 2023 are requested in month one, because January to April alone gives only a few SWOT passes.

14 Risks, Ethics and Safety

The biggest risk, SWOT not seeing the tanks, was checked in October 2026 (see Feasibility Check): SWOT sees all 13 study tanks twice every 21 days. The main risks now are "suspect" readings on village tanks, and water hidden by plants and clouds.

Table 9. Risks and mitigations

Risk	Mitigation
SWOT passes rarely, or not at all, over the chosen tanks	Checked: passes 273 and 092 both cover all 13 tanks, with 47–90 usable readings per tank since July 2023
Small tanks missing from SWOT's lake product	Checked: all 13 study tanks have a lake ID, though only 58% of mapped tanks of 6.25 ha or more do; adding tanks without one needs the pixel cloud (stretch)

Risk	Mitigation
SWOT flags almost all village-tank readings as "suspect", and a few are metres off	Remove outliers; use the five gauged reservoirs, which have both kinds of reading, to measure how good "suspect" readings are, and train the trust screen that decides which to keep (RQ2); report village tanks separately
The storage model learns from reservoirs bigger than village tanks	Use only inputs a village tank has; report tanks under 2 km² separately; check it at Kanhindawewa, and against SWOT height and AI water area at the village tanks
A chosen village tank turns out unsuitable (another cascade, no access or no gauge post)	41 other village tanks pass the same SWOT checks; swap one in
No access to official gauge records	Save the Irrigation Department's Reservoir Status data every day (automated since October 2026); request daily levels since July 2023 through the supervisor in the first month; photograph gauge posts on pass dates
Floating plants hide water: three of the eight village tanks are mostly plant-covered	Separate water-plant class plus the tank's history; report accuracy for these tanks separately; Sentinel-1 alone can mistake dense plant cover for land, so clear Sentinel-2 dates anchor these tanks; fallback: SWOT height with Imbulana's (2022) depth–area–capacity equation
Monsoon clouds	Sentinel-1 radar as a core input; Sentinel-2 used when clear (only 17–22% of dates were clear over the test tanks)
Field safety	Visit in daylight and in pairs, read gauges from the bund, never enter the water; dry-zone tanks can have elephants and crocodiles nearby
Permission	Inform the irrigation office and farmer organisation before any visit
Data volume	Process in Google Earth Engine and through the Hydrocron API; download only the study area; use Sen1Floods11 and one part of S1S2-Water, not all 165 GB

References

Sri Lankan context

Daily FT (2018). Sri Lanka designated as a globally important agricultural heritage site.

Daily FT (2023). Total water capacity of irrigation reservoirs estimated at 75%.

IFAD (2026). Water surface maps to manage ancient tank cascade systems in Sri Lanka.

Irrigation Department of Sri Lanka (2026). Reservoir Status: daily water level and available storage of major and medium reservoirs. Accessed 26 October 2026.

Madumalsha, C., Nawoda, I., Neluwala, P., Revel, M. (2025). Estimation of water elevation using ICESat-2 satellite. *Engineer: Journal of the Institution of Engineers, Sri Lanka*, 58(1).

Perera, H. E., Jayaweera, H. H. E. (2020). Use of Google Earth Engine to monitor surface water: a case study in water tanks located in the dry zone of Sri Lanka. Faculty of Science, University of Kelaniya.

University of Moratuwa studies

Dawalagala, H. S., Radampola, A., Gowsigan, P. T., Chaminda, S. P., Dassanayake, S. M., Jayawardena, C. L. (2023). Comparative analysis of water hyacinth dynamics in North Bolgoda

Lake, Sri Lanka.

Jayawardena, C. L., Brinthan, K., Thanujan, T. (2021). Evaluation of machine learning algorithms in classifying multispectral imagery on waterbody extraction.

Madhurshan, R., Dhananjaya, K. I. K., Keerthi, T., Dassanayake, S. M., Chao, Z. (2024). Multi sensor satellite remote sensing data based comparative assessment of the surface water level dynamics of the Victoria reservoir, Sri Lanka.

Meyen, A. N. (2018). Predicting reservoir water levels using deep learning. MSc thesis.

University of Moratuwa (2024). ICITR 2024 abstracts.

SWOT and related methods

AVISO. SWOT science orbit.

Bonafilia, D., Tellman, B., Anderson, T., Issenberg, E. (2020). Sen1Floods11: a georeferenced dataset to train and test deep learning flood algorithms for Sentinel-1. CVPR Workshops 2020, 835–845.

Das, P., Hossain, F. (2025). Multi-satellite tracking of surface water storage change in the era of the Surface Water and Ocean Topography (SWOT) satellite mission. *Earth and Space Science*, 12, e2024EA004178.

Fjørtoft, R., Pottier, C., on behalf of the SWOT HR Cal/Val Team (2025). SWOT HR lake product validation (slides). SWOT Science Team Meeting, Arcachon, France, 15 October 2025.

Imbulana, K. A. U. S. (2022). Depth-area-storage capacity relationships of village tanks. Engineer: Journal of the Institution of Engineers, Sri Lanka, 55(2), 53–61.

Köhn, S. J., Nielsen, K. (2026). Evaluation of SWOT HR PIXC version D water level time series of small lakes. Hydrology and Earth System Sciences, 30(20), 6443–6458.

Lappicy, T., Beltrão, D., Sales, L. O., Almeida, T., Pessoa, G. G., Souza, S., Frasson, R. P. de M., Cicerelli, R. E. (2026). A large-scale evaluation of SWOT-derived water surface elevations: precision drivers and strategies to enhance data availability. *Remote Sensing*, 18(10), 1609.

Levenson, E., Cooley, S., Webb, E., Kluetmeier, C. (2025). Improved surface water storage monitoring through multi-sensor harmonization (slides). SWOT Science Team Meeting, Arcachon, France, October 2025.

Ma, X., Wang, J., Gharari, S., Mizukami, N., Lettenmaier, D. P. (2026). Monitoring reservoir storage using SWOT satellite observations and a reservoir operation model. *Water Resources Research*, 62(3).

NASA ARSET (2025). SWOT training, Part 1 Q&A.

NASA JPL OPERA (2024). Dynamic Surface Water Extent from Sentinel-1 (DSWx-S1), version 1 (dataset description).

NASA PO.DAAC (2025). SWOT hydrology tools webinar.

Perera, C., Peiris, N., Rajapakse, L., Wijayaratna, N., Wijemannage, A. (2026). SLRainGrid-D05: high-resolution daily precipitation dataset for Sri Lanka derived from machine learning and satellite-gauge fusion. EGU General Assembly 2026, EGU26-20641.

Pruekthikanee, P. (2026). Multi-sensor deep learning fusion for spatiotemporal water surface monitoring in the Yasothon Province's Chi River Basin, Thailand. EGU General Assembly 2026, EGU26-4154 (abstract).

Wang, C.-H., Tseng, K.-H. (2025). Potential of SWOT mission in monitoring Taiwan surface water dynamics. ACRS 2025.

Wieland, M., Fichtner, F., Martinis, S., Groth, S., Krullikowski, C., Plank, S., Motagh, M. (2024). S1S2-Water: a global dataset for semantic segmentation of water bodies from Sentinel-1 and Sentinel-2 satellite images. IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, 17, 1084–1099.

Wu, J., et al. (2023). Combining satellite imagery and a deep learning algorithm to retrieve the water levels of small reservoirs. *Remote Sensing*, 15(24), 5740.

Appendix A. Research notes

Checked Oct 14, 2026. Three findings matter most: SWOT data degrades in heavy rain, ready-made SWOT time series exist only for tanks in its prior lake list, and a pretrained water model and official tank outlines already exist to start from.

A.1 Key facts to quote

Table A1. Key facts to quote

Fact	Figure	Source
Small tanks in Sri Lanka	About 23,000 tanks of 80 ha or less in the National Tanks Survey; about 21% non-functional	Lopane et al., 2025
Tanks with satellite water maps	More than 10,650 water bodies, monthly at 10 m, 2017 to August 2025	IFAD, 2026
Worst water plants in seasonal tanks (Anuradhapura District)	Water hyacinth, lotus and Salvinia	Munasinghe and Sundarabarathy (IWMI)
Silt	Sediment makes tanks shallower and cuts storage	Munasinghe and Sundarabarathy (IWMI)
SWOT accuracy targets	Water height within 10 cm for lakes over 1 km² (requirement) and 25 cm for lakes from 250 m × 250 m to 1 km² (goal), in open water; area within 15%	SWOT Science Team, 2024
Published reservoir levels	The Mahaweli Authority reports levels against spill, for example Victoria 188 cm below spill on 8 January 2026	Ada Derana

A.2 SWOT limits and how to handle them

SWOT is not a perfect gauge: rain, calm water and hills can spoil a pass, so the project keeps only clean passes and leans on Sentinel-1 in the monsoon.

Table A2. SWOT limits and how the project handles them

Limit	What happens	How the project handles it
Heavy rain	SWOT's Ka-band radar is more sensitive to rain than the Ku-band radars on most earlier altimeters, and heavy rain degrades the data; the ocean product has a rain flag, the lake product has none (AVISO)	Use quality flags and skip passes on heavy-rain days (from rainfall data); fit curves from dry-season passes; use Sentinel-1 for monsoon water area
Dark water	Calm, sheltered water reflects weakly; a known processing issue with a fix planned (SWOT Science Team, 2024)	Use quality flags; report affected tanks separately
Layover in hills	Nearly all land is in layover, though often with little impact (SWOT Science Team, 2024)	Prefer flat dry-zone cascades; treat hill reservoirs such as Victoria as a separate test
Specular ringing	Very bright reflections smear into nearby pixels	Use quality flags

Limit	What happens	How the project handles it
Only listed lakes in Hydrocron	Ready-made time series exist only for lakes in SWOT's prior lake database (NASA Earthdata)	Check each tank's lake ID first; use the pixel cloud for tanks that are missing (stretch)

A.3 Ready-made starting points

The team does not start from zero: a pretrained water model, labelled tiles, a global storage dataset and official tank outlines already exist.

Table A3. Ready-made starting points

Resource	What it gives	Use in the project
Prithvi-EO-2.0 Sen1Floods11 model	IBM and NASA water and flood segmentation model using six Sentinel-2 bands; Apache-2.0 licence	Starting model and baseline for the AI water map
Sen1Floods11	446 hand-labelled 512 × 512 tiles at 10 m, Sentinel-1 and Sentinel-2	Pre-training
GloLakes	Storage time series for more than 170,000 lakes (1984–2020), with near-real-time monitoring of 27,000; lakes over 1 km² only; volume from a terrain-slope model, not bathymetry	Comparison for the major reservoirs
UNDP tank outlines	Satellite-traced tank outlines used by Lopane et al., 2025	Tank inventory; request from UNDP
Hydrocron	SWOT time series as CSV or GeoJSON, by lake ID	SWOT water heights

A.4 Partners and data to request

Ask through the supervisor in the first month, because official records are the slowest part to get.

Table A4. Partners and data to request

Who	Ask for
Irrigation Department	Daily water levels and capacity tables for major and medium reservoirs in the study area
Mahaweli Authority	Reservoir levels against spill for the Mahaweli reservoirs
Department of Agrarian Development	Minor tank lists, gauge posts and farmer organisation contacts
UNDP Sri Lanka	The satellite-traced tank outlines
IWMI (based in Colombo)	Advice on tank cascades; possible co-supervision

A.5 Checks before the proposal

- ☐ Query Hydrocron for the candidate tanks' lake IDs and pass dates
- ☐ Count SWOT passes per 21-day cycle over Anuradhapura District
- ☐ Check which Sri Lankan reservoirs appear in GloLakes
- ☐ Request reservoir level records through the supervisor

- ☐ Run the Prithvi water model on a few Sri Lankan tanks to see where it fails

Sources

Lopane et al. (2025). Optimising water storage for climate resilience: geospatial targeting for small tanks rejuvenation in Sri Lanka. *Journal of Environmental Management*, 374.

IFAD (2026). Water surface maps to manage ancient tank cascade systems in Sri Lanka.

Munasinghe, D., Sundarabarathy, T. V. Utilization of aquatic plants. IWMI.

SWOT Science Team (2024). HR requirements status and plan.

AVISO. Winds, waves and rain seen by SWOT and nadir altimeters.

NASA Earthdata (2024). Hydrocron time series API for SWOT hydrology data.

IBM and NASA. Prithvi-EO-2.0-300M-TL-Sen1Floods11.

Cloud to Street. Sen1Floods11.

Hou, J., et al. (2024). GloLakes: water storage dynamics for 27 000 lakes globally from 1984 to present. *Earth System Science Data*, 16.

Ada Derana (2026). Mahaweli Authority issues update on water levels of major reservoirs.